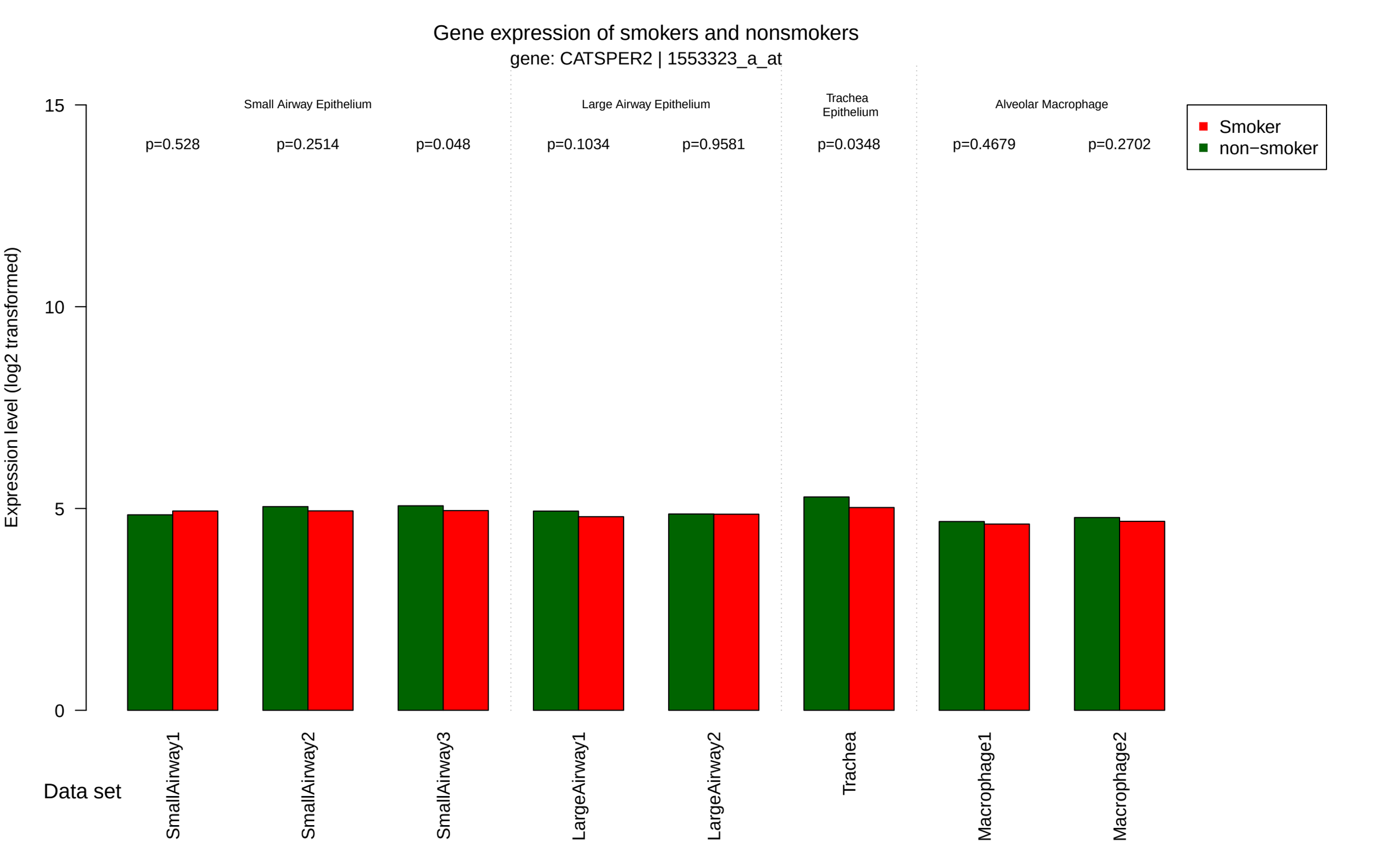
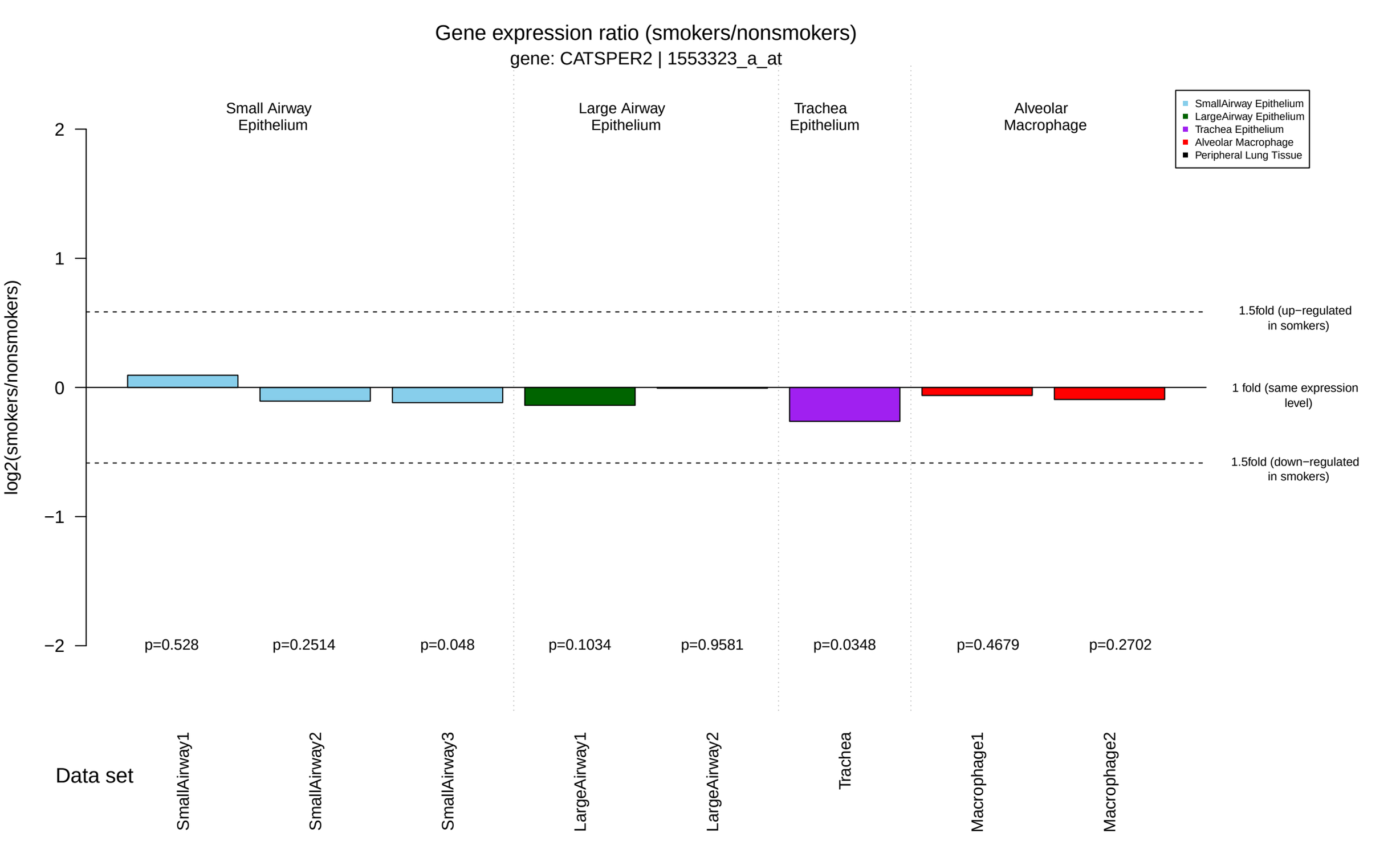
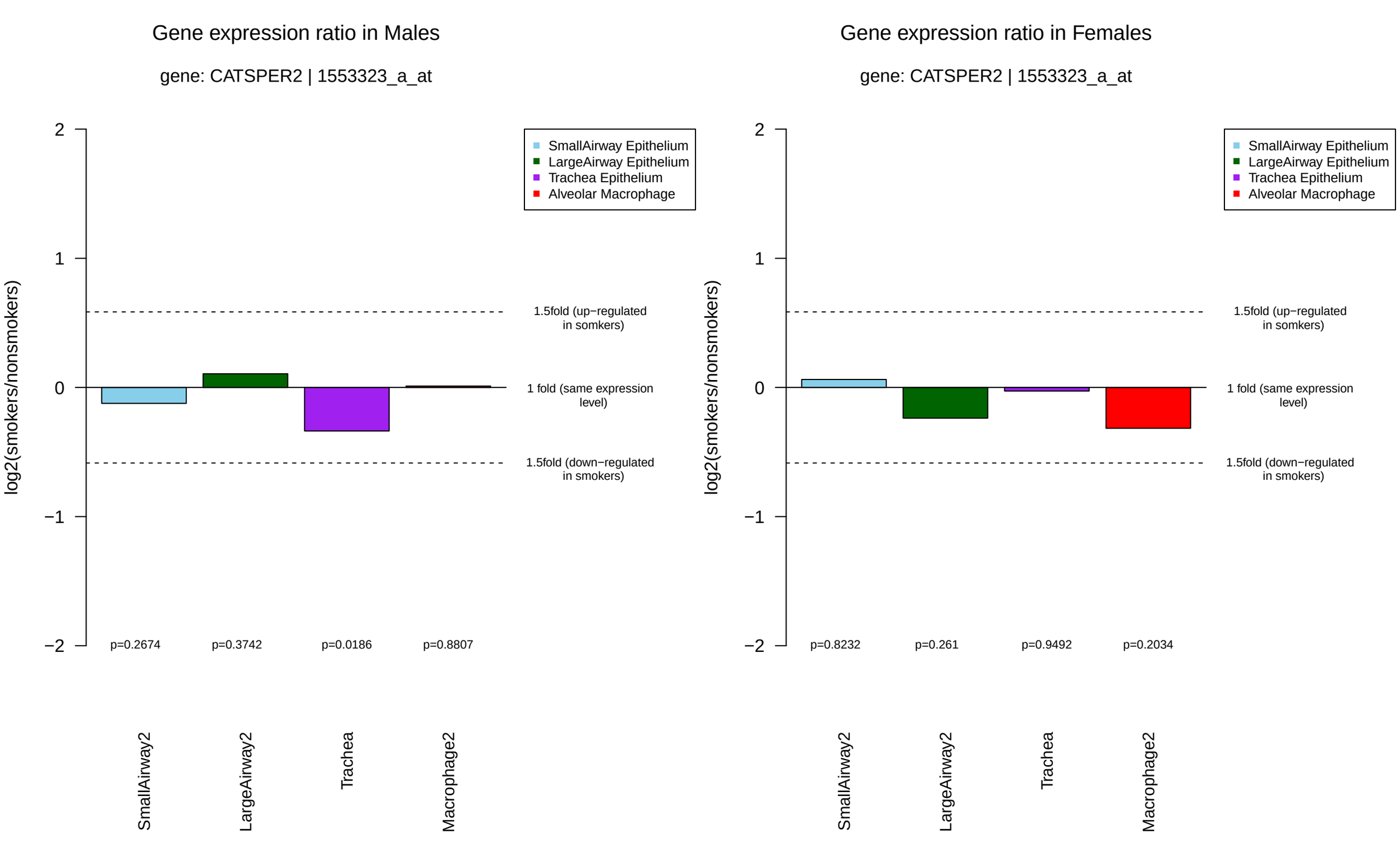
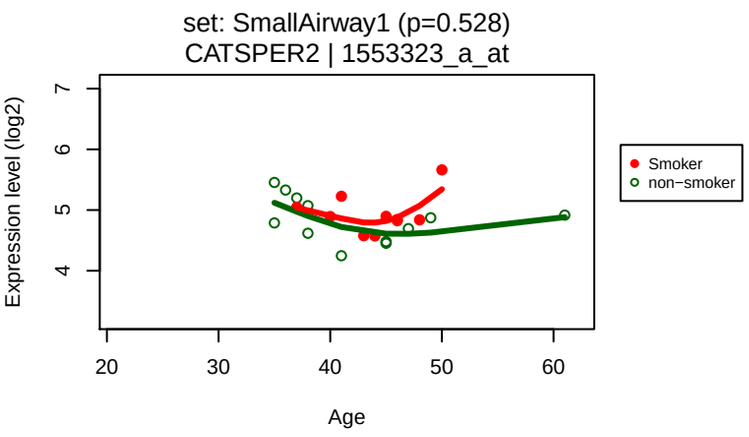
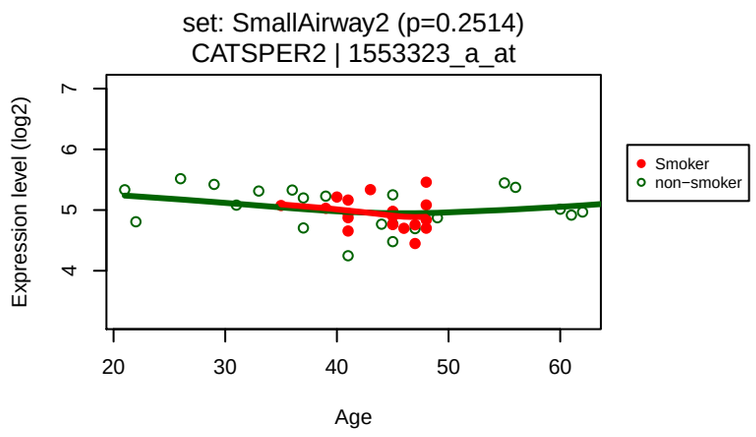
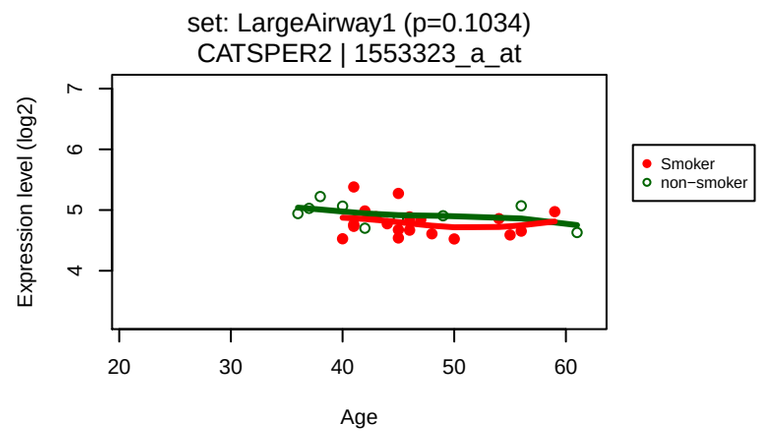

Gene expression of smokers and nonsmokers
gene: CATSPER2 | 1553323_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.528
p=0.2514
p=0.048
p=0.1034
p=0.9581
p=0.0348
p=0.4679
p=0.2702
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

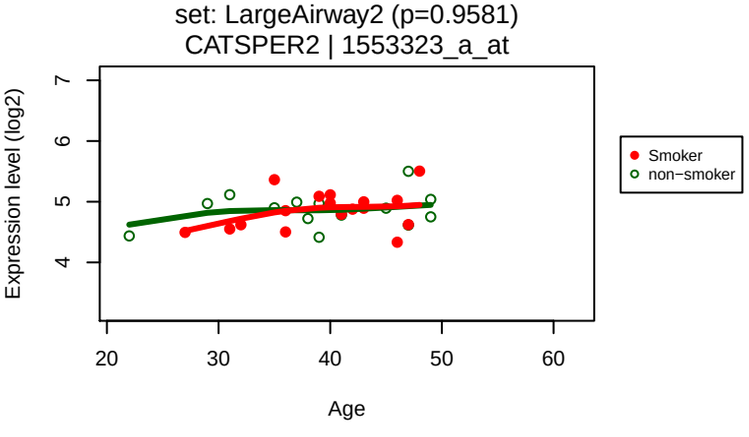

Gene expression ratio (smokers/nonsmokers)
gene: CATSPER2 | 1553323_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.528
p=0.2514
p=0.048
p=0.1034
p=0.9581
p=0.0348
p=0.4679
p=0.2702
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

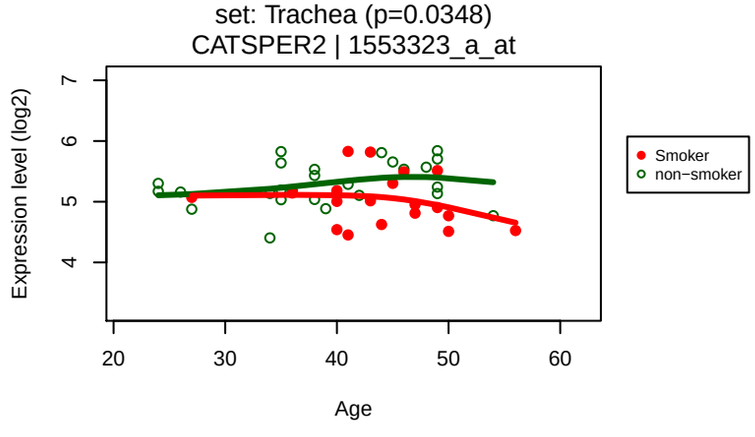

Gene expression ratio in Males
gene: CATSPER2 | 1553323_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2674
p=0.3742
p=0.0186
p=0.8807
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CATSPER2 | 1553323_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8232
p=0.261
p=0.9492
p=0.2034
SmallAirway2
LargeAirway2
Trachea
Macrophage2

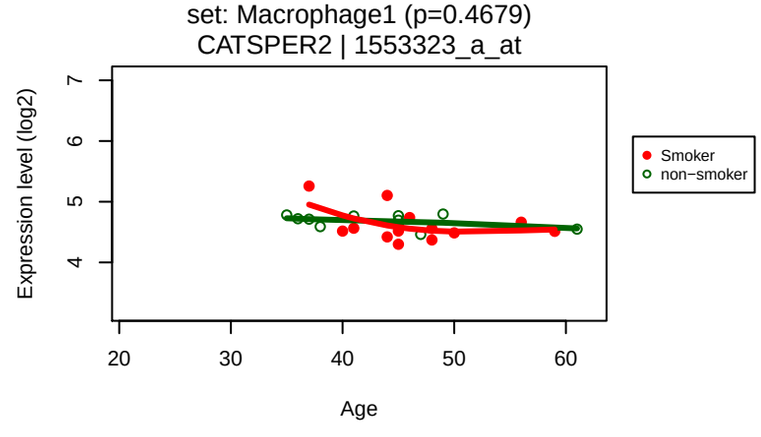

set: SmallAirway1 (p=0.528)
CATSPER2 | 1553323_a_at
Expression level (log2)
Age
Smoker
non−smoker

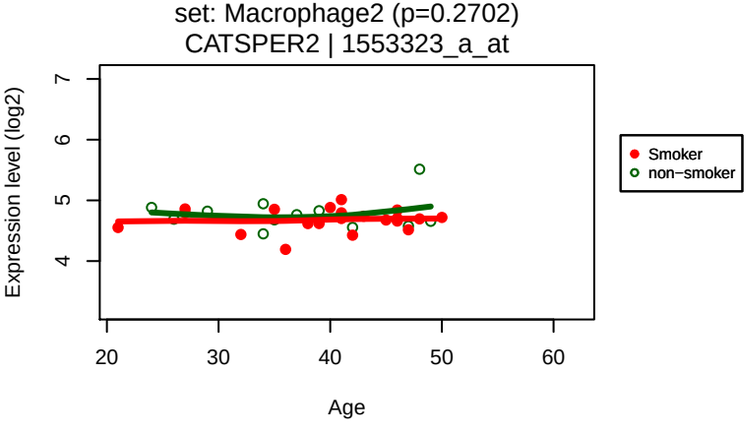

set: SmallAirway2 (p=0.2514)
CATSPER2 | 1553323_a_at
Expression level (log2)
Age
Smoker
non−smoker

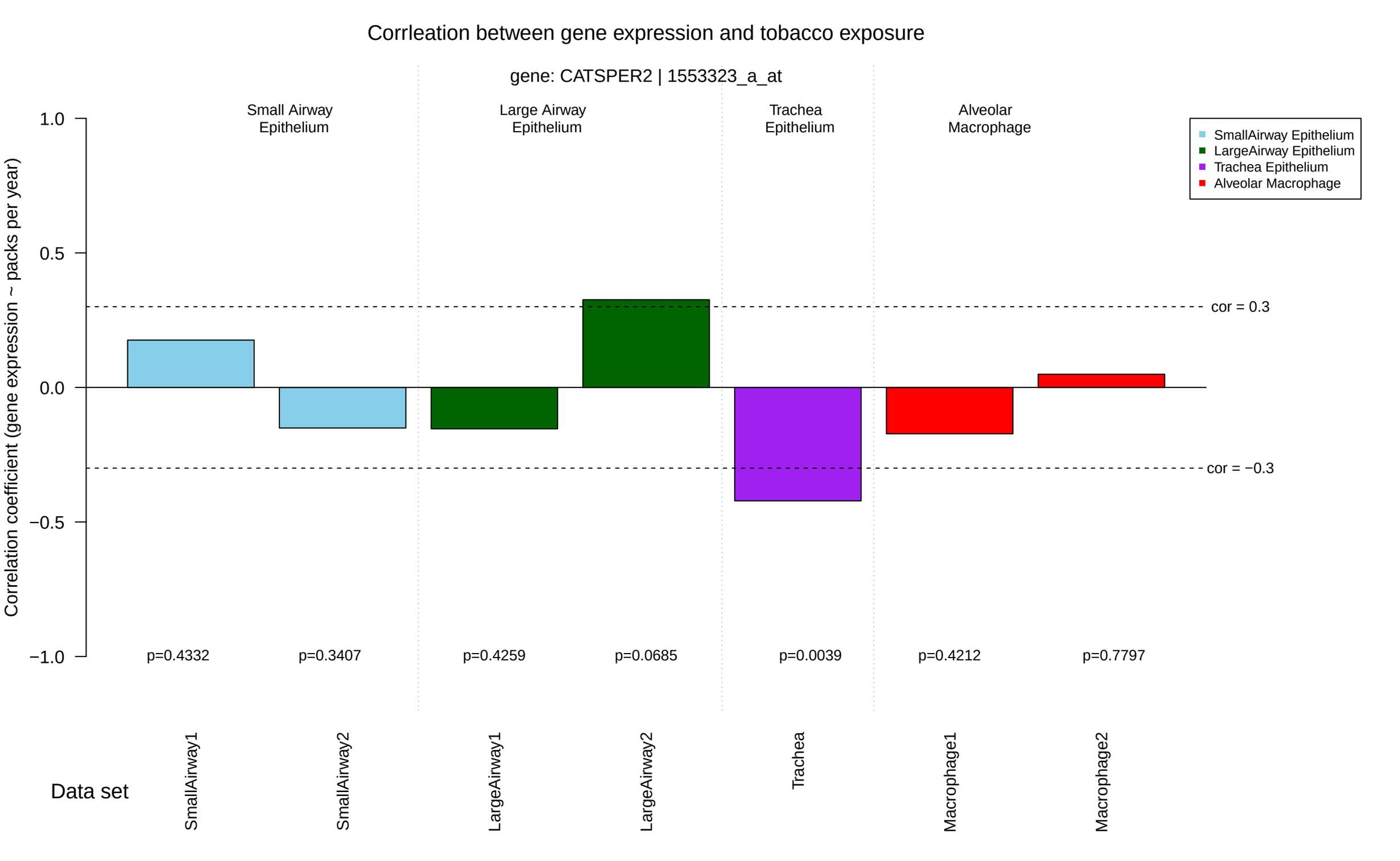

set: LargeAirway1 (p=0.1034)
CATSPER2 | 1553323_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9581)
CATSPER2 | 1553323_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0348)
CATSPER2 | 1553323_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4679)
CATSPER2 | 1553323_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2702)
CATSPER2 | 1553323_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CATSPER2 | 1553323_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4332
p=0.3407
p=0.4259
p=0.0685
p=0.0039
p=0.4212
p=0.7797
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2